In large-scale operations, the ability to swiftly reposition drones across battlespaces provides superior adaptability compared to traditional assets.

On the battlefield, drones excel in precision targeting, which lowers collateral damage and enhances mission accuracy. Loitering munitions capable of surveilling an area and striking only after confirmation of a target have become particularly valuable in operations requiring high discrimination and proportionality. UAV-assisted artillery correction and real-time strike coordination have also improved the precision of indirect fire systems, enabling engagements at longer distances while minimizing risk to friendly forces. This contributes directly to tactical lethality while preserving compliance with engagement protocols.

Moreover, drones enhance force protection by allowing militaries to conduct reconnaissance and offensive operations without exposing personnel to hostile environments. Surveillance drones monitor improvised explosive device (IED) threats, patrol conflict zones, and provide overwatch for ground units during convoy and urban operations. Maritime drones support naval forces by detecting submarine movements, monitoring coastlines, and conducting anti-piracy patrols. However, these advantages require ongoing investments in cybersecurity, communication redundancy, maintenance infrastructure, and AI reliability, as operational efficiency depends heavily on system integrity.

8.3.3 Ethical and legal considerations in autonomous warfare

While UAVs offer clear operational benefits, they also raise deeply complex ethical and legal dilemmas, particularly when autonomy reduces human control in lethal decision-making. Traditional laws of armed conflict are grounded in human judgment, accountability, and moral agency. Autonomous drones challenge this framework by allowing algorithms to influence or execute engagement decisions. This gives rise to the concern